

Shubham Sinha

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[LinkedIn](#), [GitHub](#), [LeetCode](#), [Intro video](#)

Summary

Python engineer with 5+ years of experience in backend systems, search, and applied AI. At Numerator, made analytics on a billion-row database **50%** faster and cut search updates from overnight to under a minute. Builds AI agents that investigate problems and wait for a person to approve risky actions.

Work Experience

Numerator

Jan 2024 – Present

Engineer 1

Remote, India

- Worked with the data engineering team to speed up reporting queries on a billion-row MySQL database. Heavy reports that took minutes now finish in seconds, and overall throughput is up **50%**.
- Redesigned a Django REST service on top of Elasticsearch so that business APIs, search, and data ingestion run separately. This cut median API response time by 150ms.
- Replaced the overnight rebuild of the search index with change data capture from MySQL, using pub-sub and Celery. New data now shows up in search within **60** seconds.
- Wrote Terraform to set up EC2, RDS, S3, and Lambda. Creating a new environment now takes minutes instead of hours.
- Fine-tuned Hugging Face models and cleaned product text with spaCy for the US, Canada, and Germany catalogs. Brand and category matching with any real product improved by **23%**.

Inexture Solutions LLP

Dec 2020 – Dec 2023

Python Developer

Ahmedabad, Gujarat, India

- Built UPI deposits inside a crypto app with more than **19M** registered users, so people could fund their wallets without a card.
- Set up Grafana and ELK dashboards for payments independently. Problems were spotted earlier, which cut MTTR, the mean time to recovery, by **40%**.
- Used RabbitMQ to separate services for a leading job portal, so one slow service no longer held up the others.
- Automated multi-stage builds and releases with Jenkins CI/CD, which removed most manual deploy steps.

Skills

Languages: Python, SQL, Go

Backend & Data: Django, Django REST Framework, FastAPI, MySQL, PostgreSQL, MongoDB, Elasticsearch, Celery, RabbitMQ, Kafka, Debezium, SQLAlchemy

Cloud & Delivery: AWS, Terraform, Docker, Kubernetes, Jenkins, GitHub Actions, Grafana, Prometheus, ELK

AI / NLP: LangGraph, LangChain, Hugging Face Transformers, spaCy, RAG, vector search

Projects

Incident Commander

Python, FastAPI, LangGraph, LangChain, PostgreSQL, SSE

- An assistant for on-call engineers, built with LangGraph. When an alert fires, **4** agents check logs, metrics, recent changes, and runbooks. A planner combines their findings into a diagnosis before any risky action is taken.
- Human-in-the-loop approval is required before the next step. Progress is saved in Postgres checkpoints, so an approved run continues without repeating the investigation. Updates stream live over SSE, and PagerDuty and Datadog webhooks start a run.

PulseSearch

Python, FastAPI, Debezium, Kafka, Elasticsearch, RAG, Kubernetes

- Streams MySQL changes through Debezium and Kafka into Elasticsearch. Hybrid retrieval combines keyword and vector search, and RAG answers from Groq use only the retrieved results.
- Indexing is safe to replay, using version checks and a dead letter queue for bad events. Search sync dropped from an overnight run of about 12 hours to under **60** seconds. The service runs on Kubernetes, and GitHub Actions publishes its images to GHCR.

Education

Dharmsinh Desai University

Nadiad, Gujarat, India

Bachelor of Technology in Information Technology (IT)

Aug 2017 – May 2021